

Candidate Number _____

Fraction _____

2019 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

Practice A in the **3rd** Examination Room

Zoological Experiments

All equipment and materials required for this examination have been placed on the table. Please check the items listed on the checklist. If anything is missing, please raise your hand immediately to report it to the examiner.

Equipment 1 (Please leave it on your desk during scene)		Equipment Two (One set per person, no reuse) Name	
changes) Name	Quantity		quantity
Dry bath (shared area),	3 units,	Rubber gloves,	1 pair
dissecting microscope,	1 unit,	face mask,	8, 3, 1
pointed forceps,	2 pieces,	insect needle (in a scintillation counter), red, green,	piece
pointed scissors	1	and white pins (inserted into a tech sponge), tech sponge (for use as a	3, 1
tissues	bundle,	dissection tray), cricket (in a 15 mL	tube
Cleaning	1 pack,	centrifuge tube) , water (300 μ L, in a centrifuge	2, 1
cloth, oil-based pen,	1 strip,	tube), dropper (3 mL), reagent A (in a centrifuge	tube
micro-drip straw (P200), straw	1 piece,	tube, labeled "A"),	1, 1
tip (P200), timer.	1 piece,	reagent B (in a centrifuge tube, labeled "B"), reagent C (in a centrifuge tube,	tube
	1 box, 1 piece,	labeled "C"), reagent D (in a centrifuge tube, labeled "D"), reagent E (in a	1, 1
		centrifuge tube, labeled "E"), test solution (T): Benson's solution 3 mL (in a 5	tube
		mL centrifuge tube).	1, 1
			tube 1

Important Notes 1.

Materials, medicines, and equipment on the table will not be replenished after use. 2. This exam paper

(including cover and question paper) consists of 4 pages and must be returned in its entirety upon submission.

3. This exam includes Practical Exercise A and Practical Exercise B, with a total time limit of 90 minutes.

4. Please answer on this paper. Answers may be written on the back of the exam paper, but please indicate the question number. 5. Please

abide by the basic laboratory rules during the experiment; violations will result in no points being awarded. 6.

During the experiment, you will be working in the shared equipment area. Please do not talk or make any marks other than those for answering questions

in the shared area. 7. Please be careful when using the equipment during the experiment. If you are injured, please raise your hand.

Question 1: Sex and Digestive Tract Observation of Crickets

A. Crickets have a closed, unidirectional digestive tract. Food enters through the mouth, undergoes different digestive processes in the scolex, thorax, and abdomen, and is then expelled through the anus. Salivary glands secrete saliva, which travels through the salivary ducts to the mouth, mixes with food, and then enters the esophagus to begin the digestive process. The first segment of the digestive tract is called the foregut or stomodaeum, and its main function is to break down large food particles along with saliva. The esophagus and crop also store food for entry into the middle digestive tract. True enzymatic digestion and nutrient absorption occur in the middle digestive tract, aided by villi. The posterior digestive tract contains undigested food particles, which mix with uric acid from the Malpighian tubules to form fecal particles. The rectum absorbs most of the water in these particles, and the dried fecal particles are expelled through the anus.

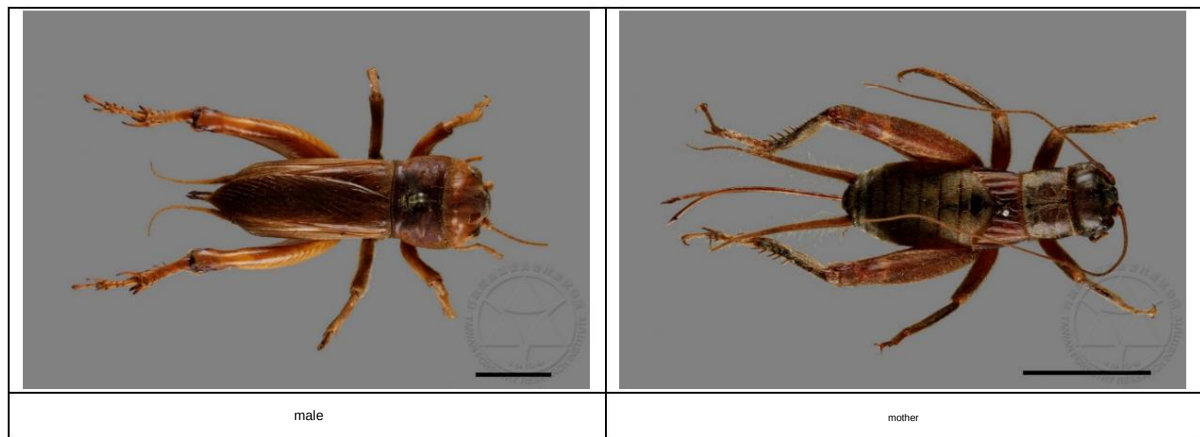
B. Commonly used methods for anesthetizing crickets include carbon dioxide and hypothermia. In this case, hypothermia was used, and the crickets were frozen beforehand.

Please note: During the dissection process, please use a scientific sponge and place the sponge under a dissecting microscope.

Avoid sticking.

C. Using the equipment and crickets provided on the table, carefully observe and answer the related questions. Cricket

reproductive system: The most significant difference between male and female crickets lies in the ovipositor at the tail.



1. Cricket Appearance

(Source: Insect Specimen Collection (Image) Digitization Project, Forestry Research Institute, Taiwan Giant Cricket)

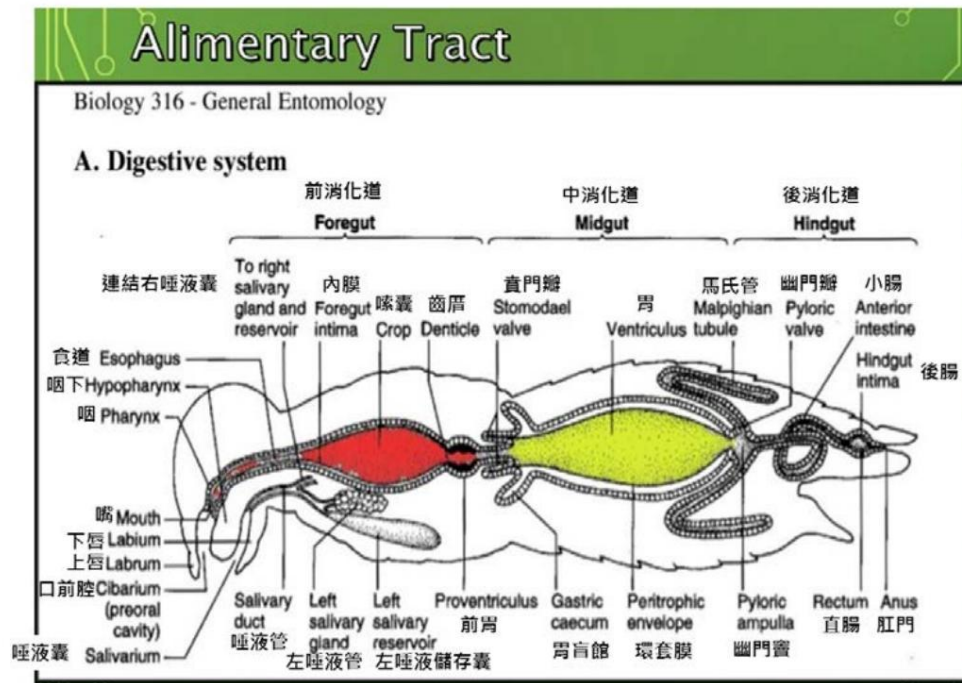


Figure 2. The digestive tract of a

cricket (Source: <https://archive.org/details/in.ernet.dli.2015.50045>, Snodgrass, 1935)

1. Please determine the sex of the assigned cricket. Mark V in the ☐ and explain your reasoning. (5 points)

(After completing the answers, please have the invigilator confirm and grade both the answers to question 2.)

☐ Female ☐ Male

Invigilator's Signature: _____

2. Using a dissecting microscope, locate the cricket's salivary sac, crop, stomach, and Malpighian tubules. Move the salivary sac to the center of the field of view, aligned with the crosshairs (and arrows), and clearly mark the locations of the crop, stomach, and Malpighian tubules using different colored pins. (20 points) (Please raise your hand

after completing the question. This answer, along with the first question, will be confirmed and graded by the invigilator.)

☐ Salivary sac

☐ Crop (Please mark with a red pin)

☐ Stomach (Please mark with a green pin)

☐ Martensitic tube (please mark with a white pin)

Invigilator's Signature: _____

3. Given that reagents A, B, C, D, and E on the table contain water, physiological saline, and starch solutions of varying concentrations,

please identify which reagents are starch solutions and determine their concentrations. (25 points) **Please** note:

1. The content must

include the experimental design, procedure, and results; otherwise, no points will be awarded. 2. Only the

provided experimental materials may be used; otherwise, no points will be awarded. 3. When

using the dry bath, please clearly label it with your exam number to avoid confusion; interference with other users' use will result in no points.

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2019 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

Practice in the 3rd Examination Room **B**

Zoological Experiments

Testing whether Vietnamese black forktail fighting fish have self-awareness

Experimental equipment:

equipment	Quantity of equipment		quantity
aquarium	1. Aquarium lid for	Vietnamese black	2
PP partition	2	forktail betta fish; 1. Central	2
plane mirror	divider; 1. Lighting	fixture.	1
tissues			1

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the question paper) has a total of 4 pages and must be returned in its entirety when submitting the exam.
3. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

Animal Group Practical Exercise – Testing Whether the Vietnamese Black Forktail Fighting Fish Possesses Self-Awareness

equipment	quantity	Equipment	quantity
aquarium	1	Aquarium Lid	2
PP partition	2	Vietnamese Black Forktail Fighting Fish	2
Flat mirror	1	Central partition	1
paper	1	Lighting fixtures	1

A. Use the mirror test to assess self-awareness.

Self-awareness is an animal's ability to observe and distinguish itself from other individuals. Understanding

Comparing which groups of animals possess self-awareness will help us understand the neuron required for cognitive abilities.

Based on physiological principles. To test whether animals possess self-awareness, Gallup, in 1970, used chimpanzees as a model.

The study participants designed a mirror test and found that chimpanzees could use their reflections to find themselves.

The markings on its body were sufficient proof that it possessed self-awareness. The same principle of detection was subsequently used in other tests.

Other groups of animals, such as birds, have also been tested, but currently only a very small number of species pass this test.

B. Facility Introduction

The experimental setup is shown in Figure 1. Both glass aquariums are equipped with lids, one side of which has a slit for operation.

A partition was used. The experimental aquarium had a plane mirror, while the control group did not. The two aquariums were separated by an opaque central partition.

The tanks are separated by partitions and illuminated by lights above. The tanks should be in the same condition as shown in Figure 2 before the experiment begins.

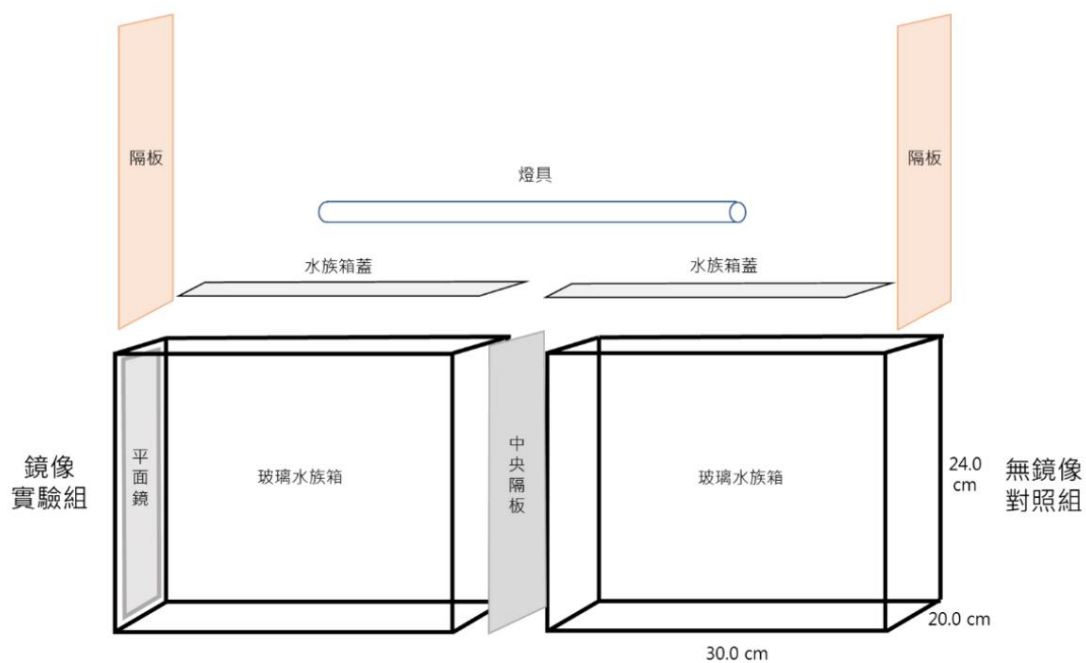


Figure 1. Schematic diagram of the main components of the experimental facility (not drawn to scale)

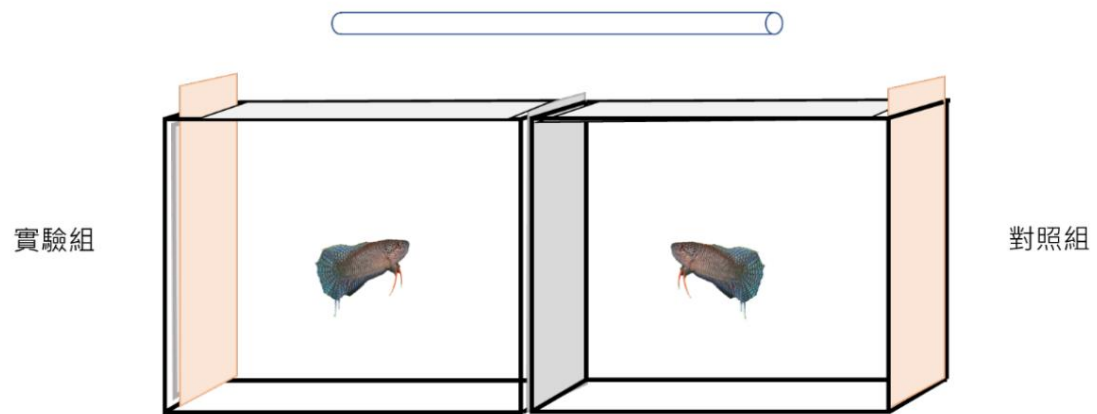


Figure 2. Initial state of the experimental facilities

C. Operating Procedures

As shown in Figure 3, the partition of the control group was first removed upwards, and the behavior of the Vietnamese black-tailed fighting fish was observed. Next, the partition of the experimental group was removed upwards, and the behavior of the Vietnamese black-tailed fighting fish in front of the mirror was observed.

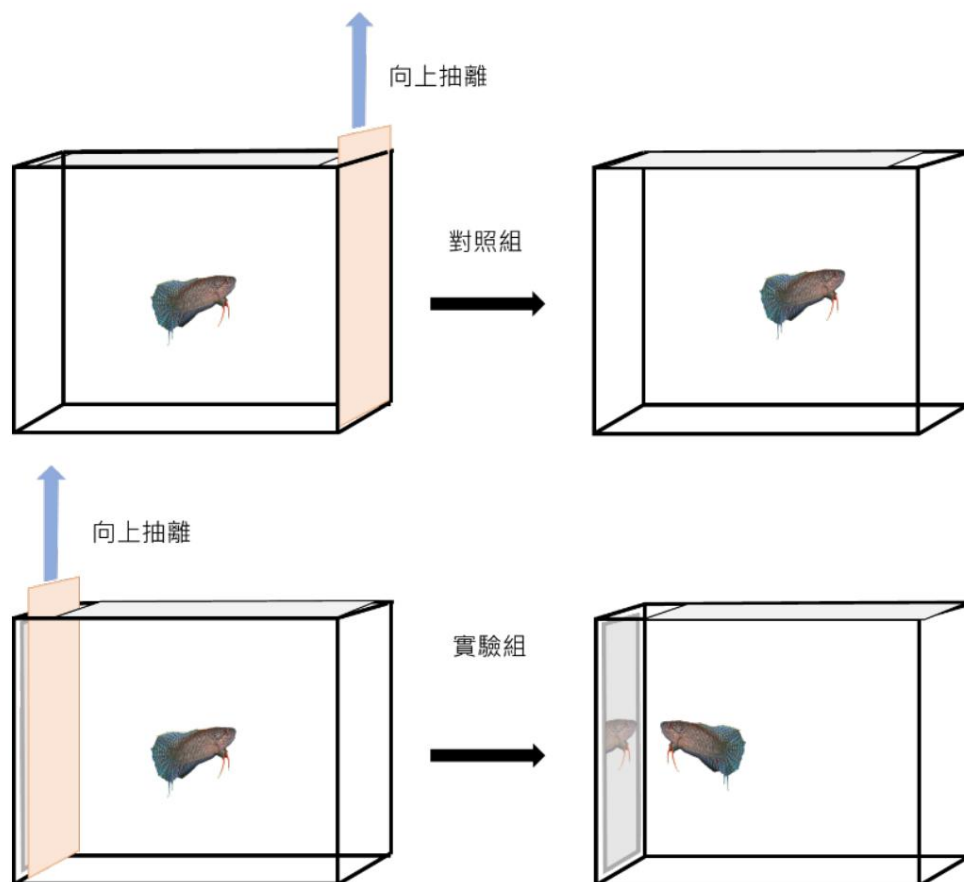


Figure 3. Experimental Operation Procedure

D. Precautions 1.

This experiment will take approximately 10 minutes.

2. Please carefully remove the partition; do not overturn the aquarium lid or bump the light fixture clips. 3. After

removing the partition, the animals may take some time to notice the mirror; please be patient. 4. To avoid disturbing the

animals, please do not move any facilities or items outside the partition, make excessive noise, or make large movements; please

remove the partition slowly and observe quietly from a distance, and do not adjust the lights.

5. After observation, please return the partition to its initial position. 6. This experiment only

needs to be performed once; it does not need to be repeated.

7. Please wipe the water droplets off the table.

E. Question

1. Animals must respond to a mirror image for the mirror test to assess self-awareness. After comparing with a control group, do

you believe the behavior of the Vietnamese black fork-tail fighting fish only occurs when a mirror image is present? What do

you think is the reason?

2. Suppose you conducted multiple replicate experiments and found the results to be consistent with what you have observed so far. What do you think of this?

Are these results sufficient to infer Douyu's self-awareness? Why?